ITM 2011

SW-API
Interfaces of the Linux Client

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Overview

- Hyperterminal
- Telnet
- Winsock Example
- Samba Share
- Projecting Files stored on the Projector
  - ➔ SPS-Interface
  - ➔ via Filename
- Virtual ComPort

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Hyperterminal

- Serial Connection to Projector
  Default-Settings:

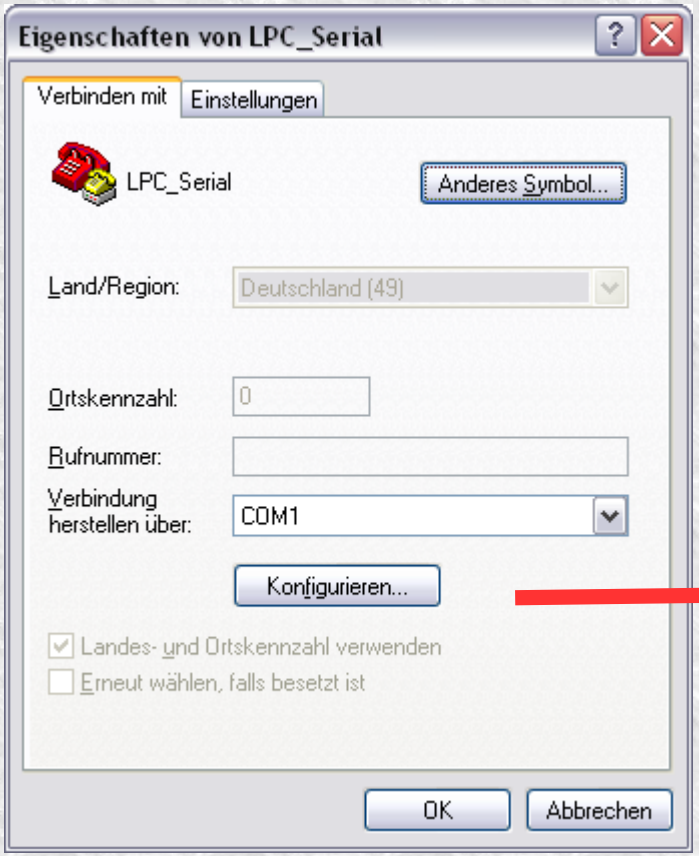
Baud	9600	(config via WebInterface)
Width	7 Bit	
Parity	Even	
Stopbits	1	
Control	None	

- Projection of HPGL-Commands
  (+ Z-Laser specific commands)
- Readable Information
  e.g. Client Version, Temperature (command `OI`)
- TimeOut, Special characters: STX, ETX
- Example

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Hyperterminal

- Hyperterminal parameters

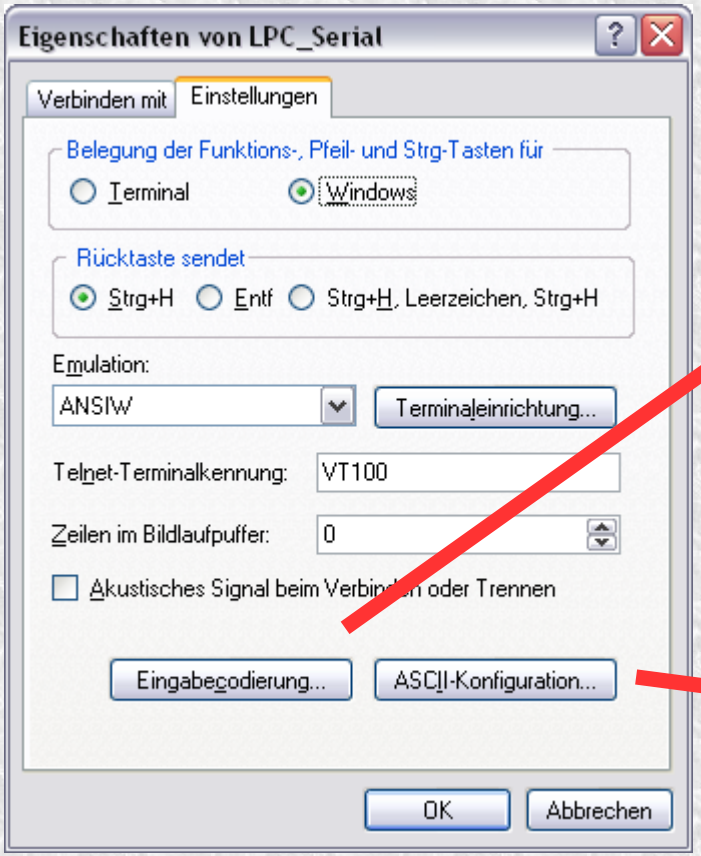


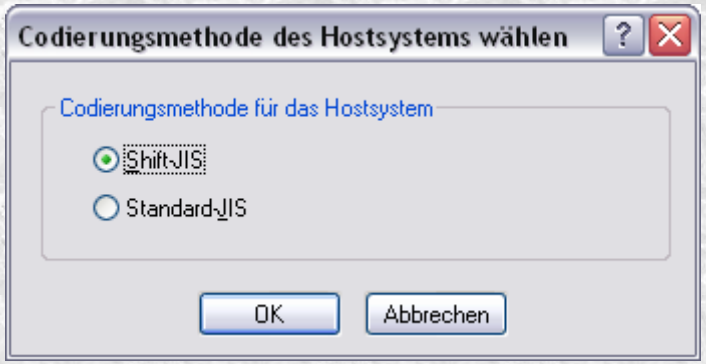


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Hyperterminal

- Hyperterminal parameters



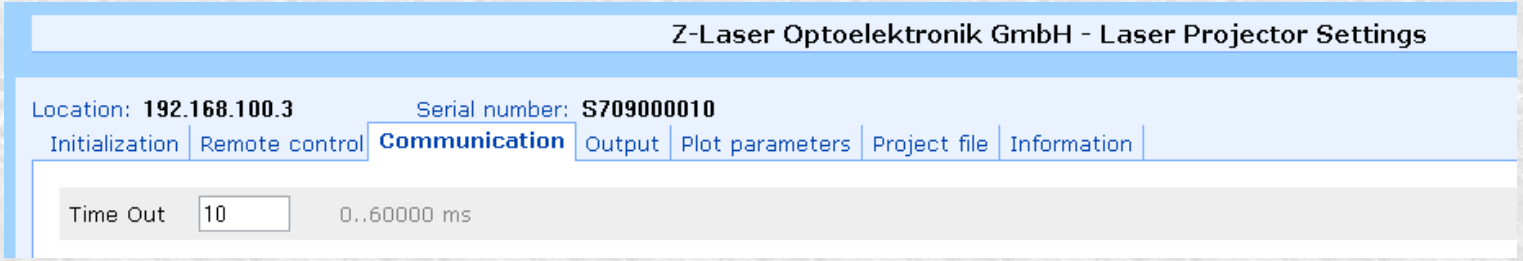




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Hyperterminal

- Parameter **`TimeOut`**



- Special Characters to "avoid" the time out:

  `STX (0x02)` → start of transmission

  `ETX (0x03)` → end of transmission

  - these are "non-printable" characters
  - to be inserted via HEX-Editor or Copy&Paste from existing file
  - "payload" has to be encapsulated between these characters
    → simply writing **`STX`** or **`ETX`** in the plt file will **not** function!
  - Z commands (e.g. ZN, ZF, ...) are ignored between **`STX`** and **`ETX`**

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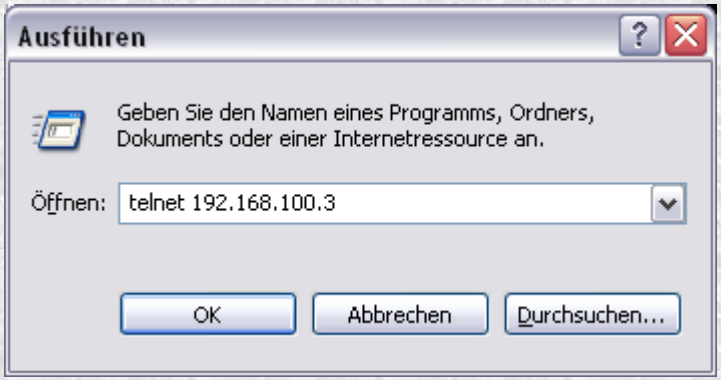
Telnet

- Network Connection to Projector via IP-Address (Standard Telnet Port = 23)
- Projection of HPGL-Commands (+ Z-Laser specific commands)
- Readable Information Client Version, Temperature
- TimeOut (`STX` and `ETX` are not available)
- Example

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Telnet

- Screenshots





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Network Folder (Samba Share)

- Purpose:
  Project/Change files, information
- Access to the folder via IP address or serial number
- Available folders on Projector:

Plot	folder used for projecting files
Data	files stored on the projector
Handbuch	LP manual (in future)
LPM	LPM setup (in future)
Info	information about the projector

- Access rights of the folders (read-only, read-write)

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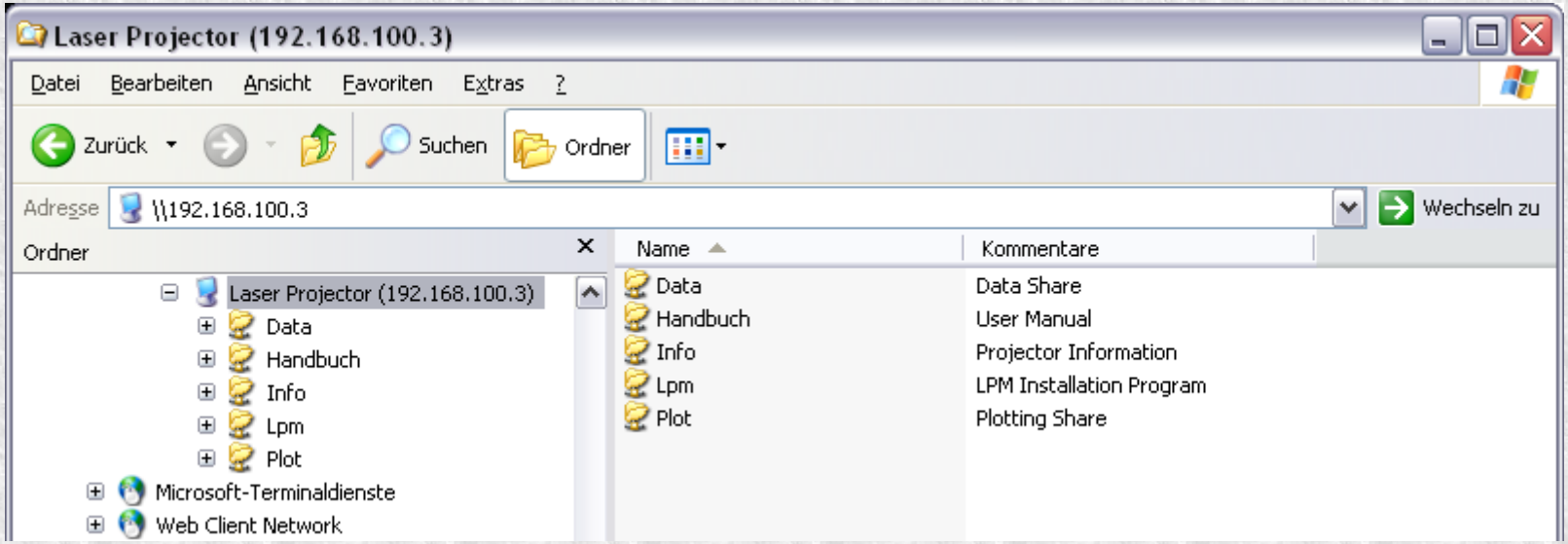
Network Folder (Samba Share)

- File System
  - 8.3 file names
    filename (w/o extension) → max. 8 characters
    extension → max. 3 characters
    longer names are “cut” to 8.3 format
  - not case sensitive
  - „safely“ store/change files by rebooting the projector
- Example: Plot folder
  (Parameter **`AutoDeletePlotFiles`**)
- the Samba share is also available in serial mode

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Network Folder (Samba Share)

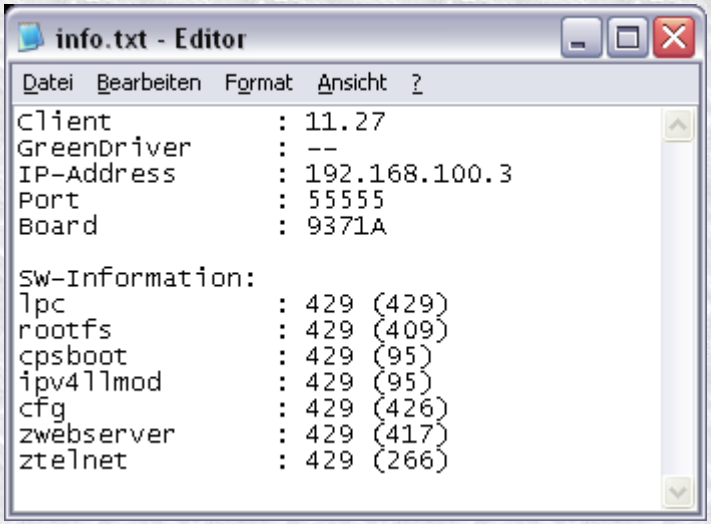
- Screenshot “Folders”



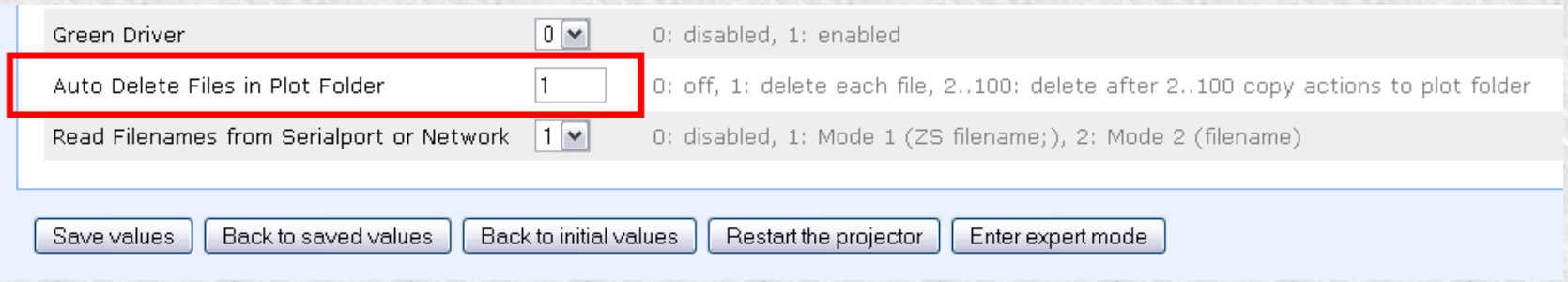
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Network Folder (Samba Share)

- Content of “info.txt” in folder “info”



- Parameter **`AutoDeletePlotFiles`**



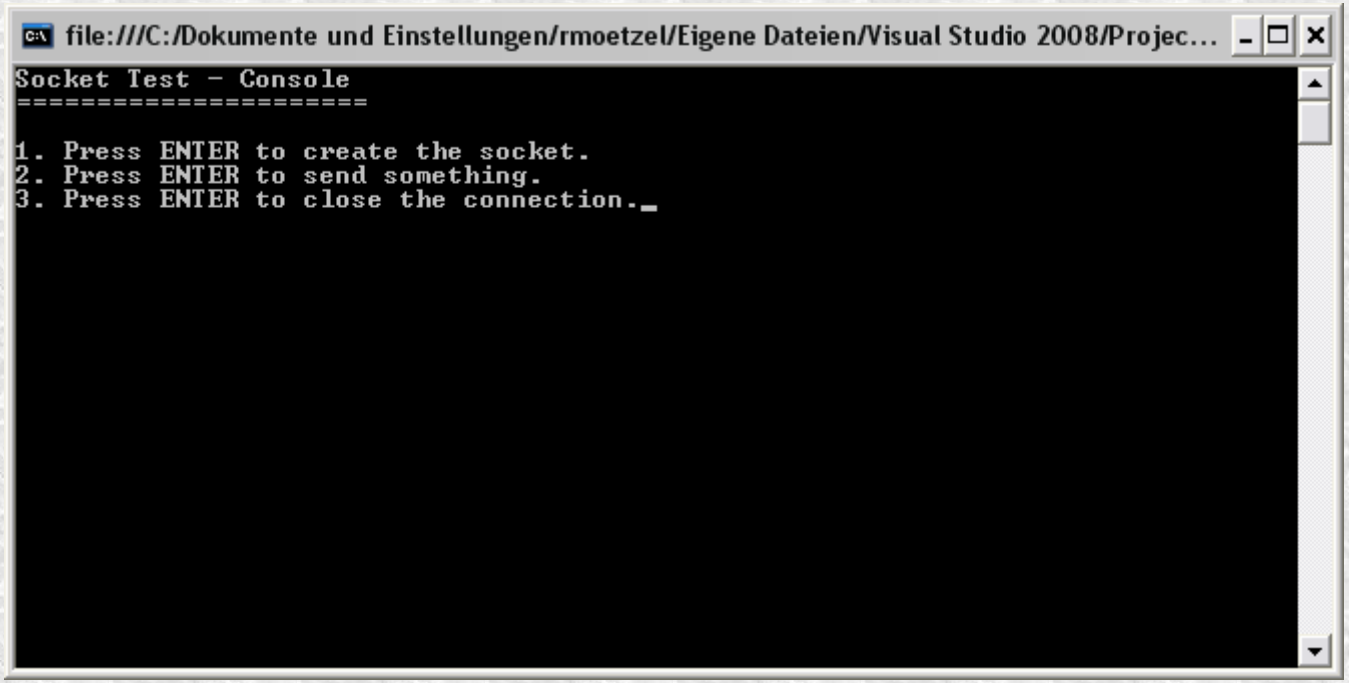
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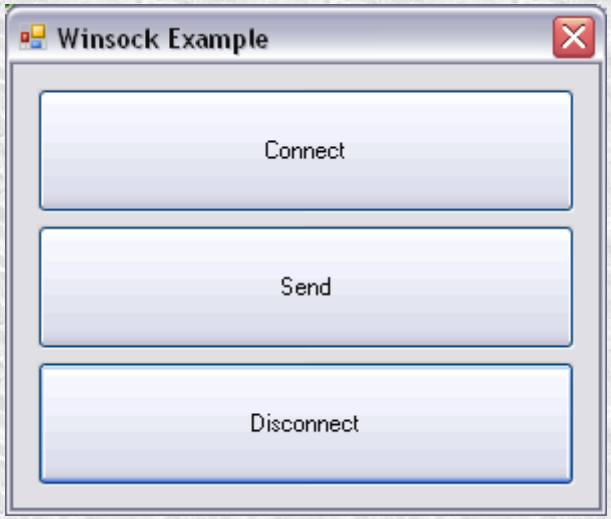
Winsock

- Principle is based on Telnet Connection
- Procedure:
  1) open socket (→ IP and port)
  2) send something (→ HPGL)
  3) close socket (→ release IP and port)
- Reaction speed depends on **`TimeOut`**
- Examples in C#
  a) Simple Console Example (Live Development)
  b) Simple GUI Example (Show)

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Winsock - Examples





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Winsock - Example-Code

```
using System;
using System.Collections.Generic;
using System.Text;
// use Sockets
using System.Net;
using System.Net.Sockets;
namespace ConsoleApplication1
{
   class Program
   {
      static void Main(string[] args)
      {
         // variables
         Socket s;
         IPEndPoint iep;

         Console.WriteLine("Socket Test - Console\n======================\n");
         // create socket
         Console.Write("1. Press ENTER to create the socket.");
         Console.ReadLine();
         s = new Socket(AddressFamily.InterNetwork, SocketType.Stream, ProtocolType.Tcp);
         iep = new IPEndPoint(IPAddress.Parse("192.168.100.13"), 23); // telnet port --> 23
         s.Connect(iep);
         // send something
         Console.Write("2. Press ENTER to send something.");
         Console.ReadLine();
         ASCIIEncoding enc = new ASCIIEncoding();
         byte[] content = enc.GetBytes("SP1; PU; PA 0,0; PD; PR 1000,0; PR 0,200; PR -1000,0; PR 0,-200;
PU;" +
            "PA 0,100; PD; SI 80,60; LB WINSOCK TEST;DU");
         s.Send(content);
         // close connection
         Console.Write("3. Press ENTER to close the connection.");
         Console.ReadLine();
         s.Close();
      }
   }
}
```

use sockets

create socket/connection

„prepare“ content

send content

close connection

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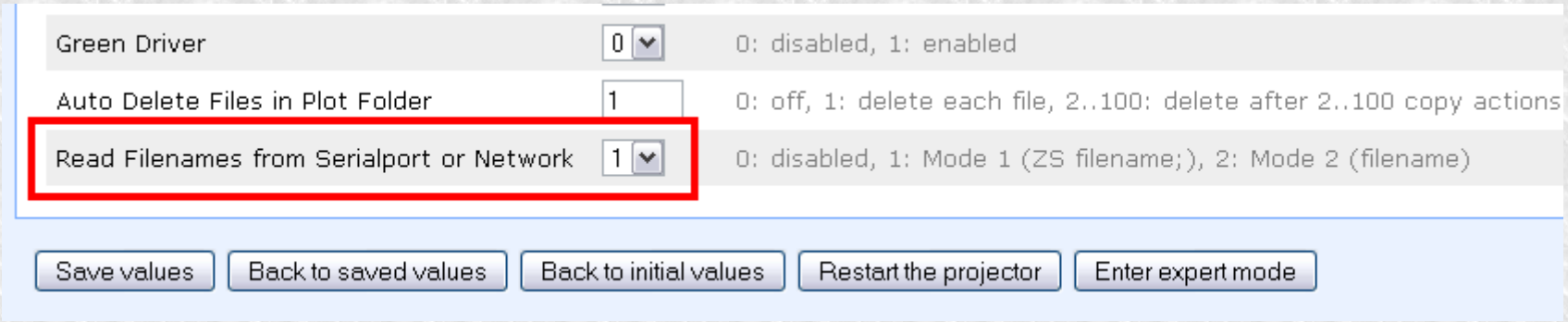
Project Files From Projector Serial

- Projector connected via Serialport
- Parameter `UseFileNameFromNetCom = 1`
- Store Files on Projector's Folder “Data”
  `\\<IP>\Data`
  or
  `\\<SerialNumber>\Data`
- Select/start projection by sending the Filename via Serialport
- Change Files
- Example

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Project Files From Projector Serial

- Parameter **`UseFileNamesFromNetCom`**



- Select projection-file by sending the following to the serialport (e.g. with Hyperterminal):

`ZS <Filename_without_ending>;`

start-marker — filename (w/o postfix) — end-marker

e.g. **`ZS serial1;`**
`ZSserial2;`

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Project Files From Projector Network

- Projector connected via Network
- Parameter `UseFileNameFromNetCom = 1`
- Store Files on Projector's Folder “Data”
  `\\<IP-Address>\Data`
  or
  `\\<SerialNumber>\Data`
- Select/start projection by sending the Filename via Network connection
- Change Files
- Example

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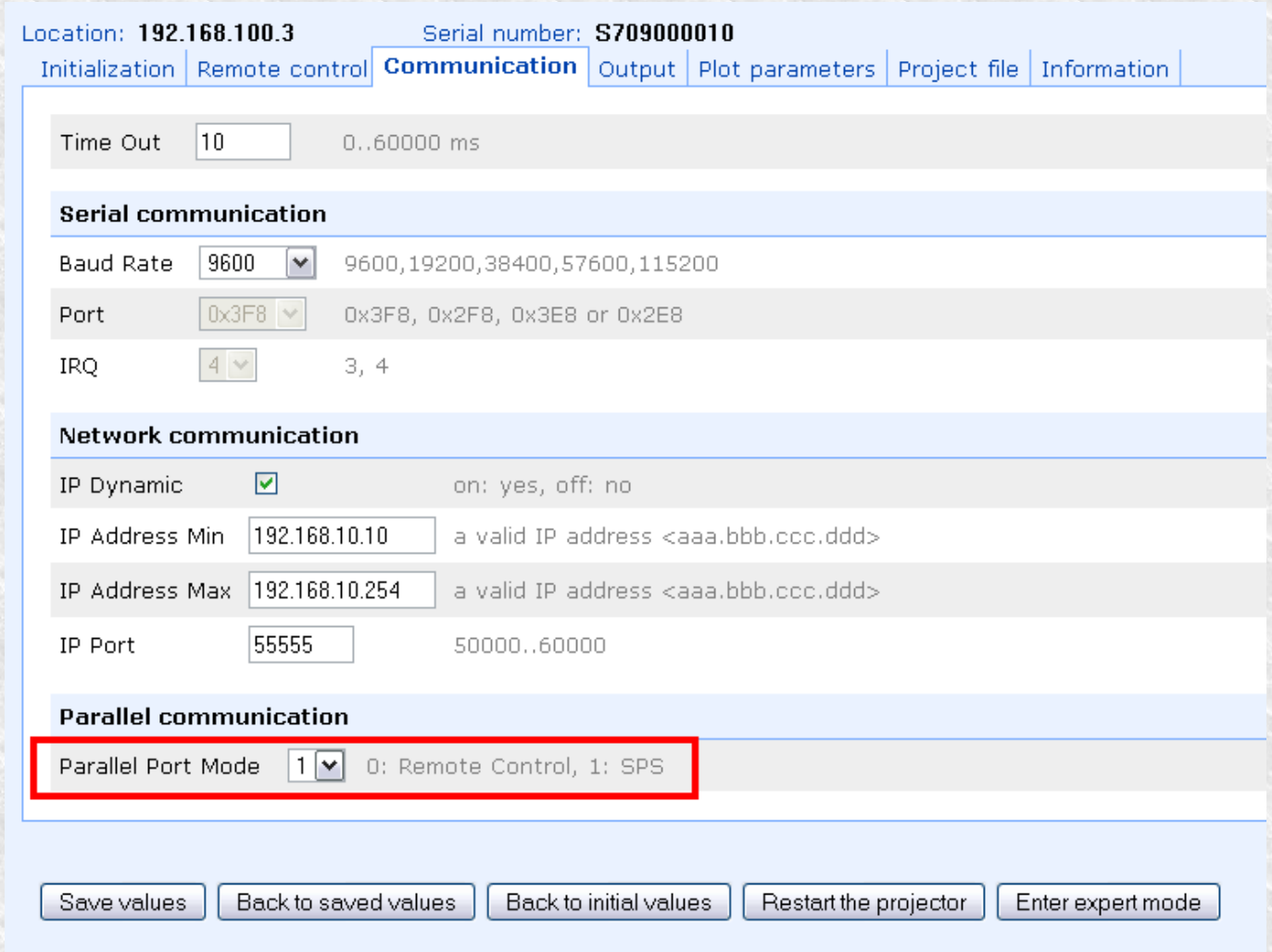
Project Files From Projector SPS-Interface

- Projector connected via SPS-Interface
  e.g. switch box or other „controller device“
- Parameter **`ParPortMode = 1`**
- Structure of the CSV file
- Store CSV Files on Projector's Folder
  `\\<IP-Address>\Data` or
  `\\<SerialNumber>\Data`
- select projection by setting bits on the SPS-Interface
- Example with Switchbox

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Project Files From Projector SPS-Interface

- Parameter **`ParPortMode`**



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Project Files From Projector SPS-Interface

- Structure of CSV file

```
<D0..D7>,"<command [parameter]>",,,
```

bit pattern — command + parameter — not used

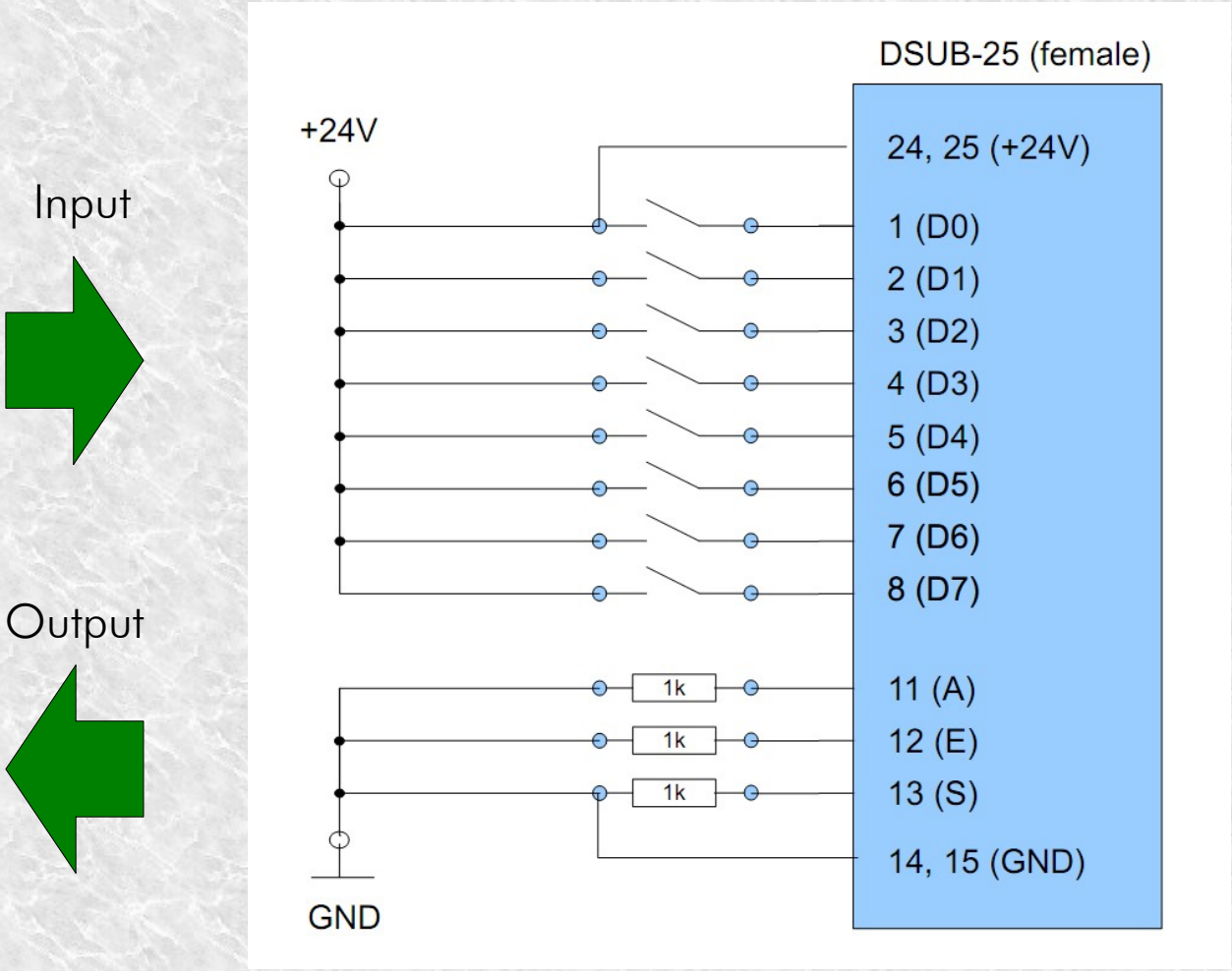
Example

```
00000001,"/FO quad.plt",,,
00000010,"/FO roof.plt",,,
00000011,"/FO xy.z",,,
00000100,"/BA",,,
00001000,"/BE",,,
00001100,"/NEXTCSV",,,
```

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Project Files From Projector SPS-Interface

- Schematic of Switchbox



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Project Files From Projector SPS-Interface - Hints

- to „safely“ store changed files on the projector do a projector restart/reboot via WebInterface
- use an existing and working CSV file as template for any new CSV file
- filenames have to be in 8.3 format
- change the CSV files with a text-editor only (do NOT use Excel)

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Virtual Com Port

- Emulate a serial port (COM-Port) via Network
- Proprietary tool from
  http://www.hw-group.com/products/hw_vsp/hw_vsp2_de.html
  (Free only for private use, see license!)
- Useful for programs supporting only serial connections, e.g.
  a) DOS-Mode-Example (COM5)

```
mode COM5:9600,e,7,1      setup COM
copy quad.plt COM5        project
```

  b) Hyperterminal
  same as “normal” serial Hyperterminal
  c) …

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